

Problem 9

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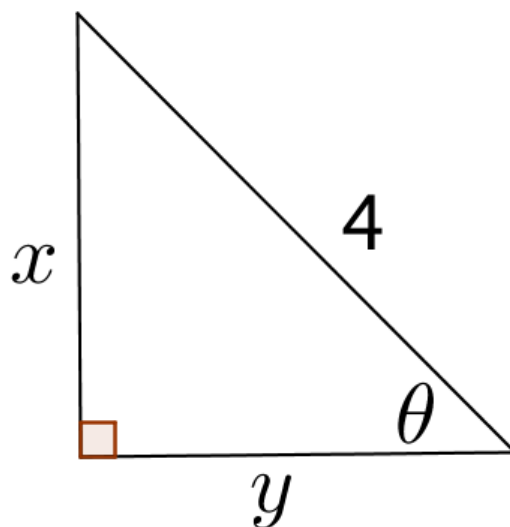
Problem 1 Simplify each of the following expressions.

(a) $\cos^{-1}(\sin(\pi/2))$

Explanation. By the unit circle, $\sin(\pi/2) = 1$, and so we are looking for $\cos^{-1}(1)$. The range of $\cos^{-1}$ is $[0, \pi]$, and so, by properties of inverse functions, $\cos^{-1}(1) = 0$.

(b) $\tan(\sin^{-1}(x/4))$

Explanation. Let $\theta = \sin^{-1}(x/4)$, then $\sin(\theta) = x/4$. We can then draw the corresponding right triangle:



Calling the adjacent side y , by the Pythagorean Theorem we obtain

$$4^2 = x^2 + y^2 \implies y = \sqrt{16 - x^2}$$

Remark: Since θ is in the range of $\sin^{-1}$, it follows that $-\pi/2 \leq \theta \leq \pi/2$. On the other hand, the expression $\tan(\sin^{-1}(x/4))$ is defined only for

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$-4 < x < 4$ (Note: $\sin^{-1}(4/4) = \frac{\pi}{2}$ and $\tan$ is not defined at $\frac{\pi}{2}$; similarly for $x = -4$.)

Therefore, $-\pi/2 < \theta < \pi/2$, which implies that $\cos(\theta) = y/4 > 0$. Therefore, $y > 0$.

Then

$$\begin{aligned}\tan(\sin^{-1}(x/4)) &= \tan(\theta), \\ &= \frac{x}{y}, \\ &= \frac{x}{\sqrt{16 - x^2}}.\end{aligned}$$

Note: $\tan(\theta)$ has the same sign as x , since $y > 0$.